

SHIVANSHU SRIVASTAVA

Varanasi, India – 221010 | +91 9198865041 | shivanshu.srivastava.work@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Backend Engineer and Python Developer focused on architecting high-performance APIs, asynchronous data pipelines, and scalable system infrastructure. Proven experience engineering institutional-grade trading backends (FastAPI, WebSockets, MongoDB), autonomous RAG systems (LangChain, ChromaDB), and resilient automation architectures. Adept at leveraging object-oriented design and async paradigms to resolve distributed computing bottlenecks, optimize database retrieval, and accelerate system deployments.

Education

Sandip University <i>B.Tech, Computer Science and Engineering (GPA: 8.49)</i>	2022 – 2026 <i>Nashik, Maharashtra</i>
GlenHill School <i>SSC & HSC</i>	2018 – 2021 <i>Varanasi, Uttar Pradesh</i>

Technical Skills

- **Languages:** Python 3.x, SQL, HTML/CSS, Bash
- **Frameworks & Libs:** FastAPI, Flask, Django, Pandas, NumPy, BeautifulSoup, Requests, LangChain
- **Databases:** PostgreSQL, SQLite, MongoDB, MySQL, SQLAlchemy ORM
- **Tools & Platforms:** Git, GitHub, VS Code, Postman, Linux/Ubuntu, Docker, Pytest
- **Core Concepts:** REST APIs, OOP, Distributed Systems, Web Scraping, AsyncIO, Vector Databases

Internships

Brain O Vision <i>Machine Learning Intern</i>	Jan 2024 – Mar 2024 <i>Hyderabad, India</i>
– Developed scalable feature engineering pipelines and behavioral clustering models (K-Means) to preprocess and segment large consumer datasets. – Fine-tuned XGBoost and Random Forest algorithms, achieving measurable improvements in predictive accuracy for customer cohort analysis. – Collaborated with cross-functional teams to integrate predictive models into analytical dashboards for marketing optimization.	
Code Alpha <i>Python Programming Intern</i>	Nov 2025 – Nov 2025 <i>Lucknow, India</i>
– Engineered asynchronous Python workflows to automate data ingestion and processing, reducing routine task execution time by over 40%. – Strengthened system reliability by implementing comprehensive Pytest suites, increasing test coverage and accelerating deployment cycles. – Refactored legacy monolith scripts into modular, object-oriented micro-components, significantly improving maintainability and system extensibility.	

Projects

TradeFlow – Trading Terminal

Live | Python, FastAPI, MongoDB, LangChain, WebSockets

- Engineered a high-performance trading API using FastAPI, implementing WebSockets for sub-100ms real-time market data streaming.
- Designed a highly scalable MongoDB schema enforcing strict per-user data isolation; integrated secure JWT-based authentication.
- Architected an AI coaching module leveraging LangChain and Groq LLMs to automate position sizing and execute risk analysis pipelines.

NSE Swing Screener

Live | Python, FastAPI, SQLite, BeautifulSoup, AsyncIO

- Architected a custom parallel processing engine scanning 500+ NSE stocks for quantitative trading signals.
- Reduced historical data analysis latency by leveraging asynchronous task scheduling (APScheduler) and optimized SQLite caching layers.
- Served processed market insights and breakout patterns via a lightweight FastAPI backend for frontend consumption.

Autonomous Org Memory (ThirdEye)

Source | Python, ChromaDB, RAG, LangChain, FastAPI

- Engineered a scalable Retrieval-Augmented Generation (RAG) backend utilizing FastAPI and ChromaDB vector store.
- Solved unstructured data retrieval bottlenecks by implementing semantic document chunking and optimized vector search.
- Enabled context-aware LLM query responses for internal knowledge bases, securely exposing answers via API endpoints.

NSE Momentum Scanner

Source | Python, BeautifulSoup, GitHub Actions, Telegram API

- Automated daily market scanning by building a resilient GitHub Actions pipeline that scrapes and processes Nifty 500 trends.
- Engineered real-time breakout alerting algorithms with instant push notifications dispatched via the Telegram API.

Hackathons

Regulatory Guideline Monitor & Updater (Top 10)

April 2026

Sunhacks 2026

- Engineered a real-time compliance dashboard tracking regulatory updates via webhooks from NSE, BSE, MCX, and RBI.
- Automated policy diffing and alerts for critical trading guideline changes.

HVAC Smart Control System (Top 10)

April 2024

Sunhacks 2024 by Sandip University & ESDS

- Designed a predictive machine learning pipeline for HVAC energy optimization and smart climate control.

Sales Prediction & Product Performance Predictor Model

Oct 2023

AICTE Data Science Hackathon (Winner)

- Secured 1st place by engineering high-accuracy predictive regression models for sales intelligence and product forecasting.